Extrapolating Venusian Atmospheric Profiles using MAGMA Gaussian Processes

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Context

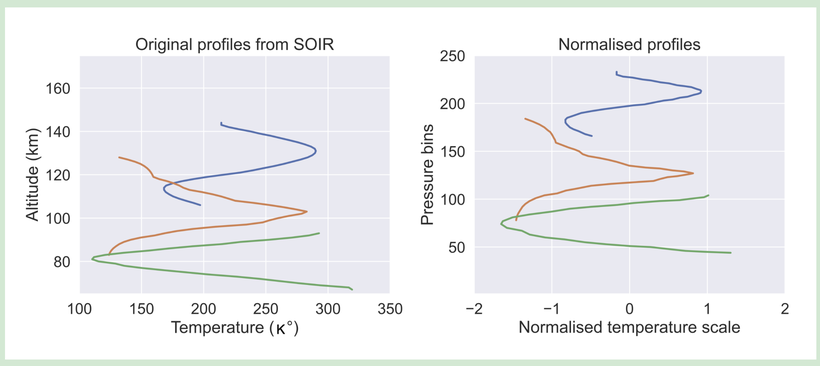
From 2006 to 2014, **Venus Express** orbited Venus. One of its instruments, **SOIR**, collected data on the temperature and composition of the atmosphere. The resulting dataset contains several hundred atmospheric profiles. However, these profiles are incomplete: in most cases, only a narrow range of 10 to 50 consecutive kilometers could be observed. Some probabilistic models, such as **Gaussian Processes** (GP), can help predict the missing altitudes by extrapolating the profiles. These models seem appropriate as they provide uncertainty estimates and can be tuned with expert knowledge, but they struggle to handle a large number of profiles to improve predictions. A novel and promising model to address these issues is **MAGMA**.

Can MAGMA be applied to an atmospheric dataset and provide better results than a classical GP?

Methodology

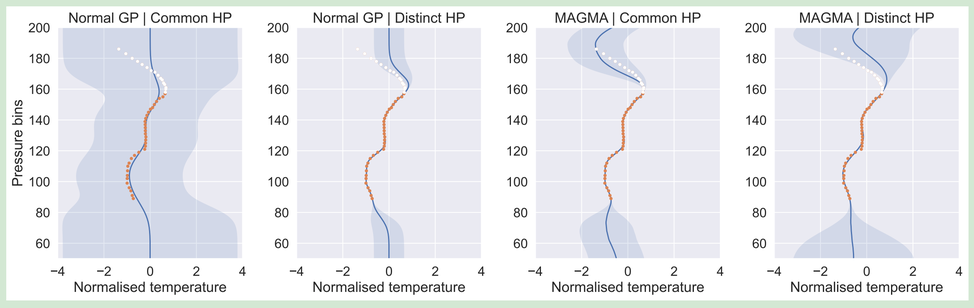
The goal of the experiment is to train both a MAGMA GP and a classical GP on the data and evaluate their predictions. The SOIR dataset is pre-processed as follows:

- Profiles are normalized, and profiles with too few observation points are removed.
- Altitude is replaced with a pressure scale.
- The dataset is divided into training and test sets.
- For profiles in the test set, one-third of the points are marked as "unseen test points."



Profiles from the training set are used by both models to learn their hyperparameters (HP). This can be done in two ways: either using common HP for every profile or distinct HP for each profile. To start making predictions, the models are given two-thirds of the points from each test profile. Their performances are then evaluated on the remaining unseen test points. These unseen points are consecutive and can be located at the top, middle, or bottom of the profile.

4 predictions from different models



Results

		Unseen test obs.	
		MSE	CIC_{95}
Baseline GP	**Distinct HP**	0.3772	81.61%
	Common HP	0.4781	100%
MAGMA	**Distinct HP**	0.2804	78.73%
	Common HP	**0.2378**	**94.67%**

The table above shows the average performance of both models in terms of precision (MSE) and the plausibility of their predicted confidence intervals (CIC_{95}). The best model is the one with the smallest MSE and a CIC_{95} closest to 95%.
Using a MAGMA GP with common hyperparameters for the whole dataset gives the best results, both in terms of prediction accuracy and the reliability of the confidence intervals.

Conclusion & Future Works

- MAGMA is a promising model for atmospheric profile extrapolation, providing good predictions and confidence intervals close to the known data.
- Multiple extensions of MAGMA can be explored to achieve more accurate predictions specific to our data.
- Probabilistic models can be applied to limited datasets from space science, providing uncertainty estimates and explainability.

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Complete paper can be accessed here: